A finite RKBS where ERM representers avoid the training points

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Abstract

Bartolucci–De Vito–Rosasco–Vigogna ask how Hilbert-space representer theorems extend to Banach spaces and later note that their RKBS representer theorem does not imply that the representing atoms are located at the training points. We give a finite three-point RKBS in which the corresponding training-site strengthening is false. For a convex coercive loss, the unique regularized empirical-risk minimizer is represented by a single atom at a point which is not in the training set.

1 Source context

The source paper is arXiv:2109.09710v2, "Understanding neural networks with reproducing kernel Banach spaces" [1]. Its introduction frames the natural Banach analogue of the classical RKHS representer theorem as follows.

> variations of the corresponding measures. The study in [2] provides a clear discussion on this perspective.
>
> The extension from a Hilbert to a Banach setting opens a number of questions. We discuss two that are relevant to our study. The first one is related to the characterization of the solution of empirical minimization problems, the so-called representer theorem. It is well known that, in a Hilbert setting, minimizers always lie in a finite dimensional subspace. Each solution is a linear combination of the reproducing kernel associated to the Hilbert space evaluated at the training set points [23, 24, 44]. This result has immediate computational implications and is at the base of kernel methods [45]. A natural question is then how these results extend to a Banach space of functions defined by neural networks. A number of recent results tackles this question [52, 35]. A main difficulty is that the Banach spaces defined by neural networks are non-reflexive, and their definition requires some care. In this context, our first contribution is that

In the RBF/kernel-mean-embedding example following the RKBS representer theorem, the authors explicitly point out that the theorem gives atoms at points $x'_i \in X$, but does not imply that those points belong to the training set.

It is interesting to note that this is exactly the kernel mean embedding of μ (see [30] and references therein). Hence $\mathcal{B}$ is a subspace of $\mathcal{H}$ and, since the kernel mean embedding is continuous from $\mathcal{M}(\Theta)$ into $\mathcal{H}$, the norm $\|\cdot\|_{\mathcal{B}}$ is stronger than the norm induced by the scalar product of $\mathcal{H}$. For example, if the kernel κ is characteristic [30], the map $\mu \mapsto f_\mu$ is injective, so that $\mathcal{B}$ is isometrically isomorphic to $\mathcal{M}(\Theta)$, which is not separable, whereas $\mathcal{H}$ is separable since $\mathcal{X}$ is. Still, Theorem 3.9 states the existence of solutions of the form

$$f = \sum_{i=1}^{K} \alpha_i K_{x_i'}, \qquad x_i' \in \mathcal{X},$$

with $K \leq N$. Note however that Theorem 3.9 does not imply that the points x_i' belong to the training set $\{x_i\}_{i=1}^N$.

The example below shows that this caveat is sharp even in a finite positive definite kernel model.

2 Idea of the proof

In an RKHS, the squared norm makes the training-site span orthogonal to the null directions of the sampling map, so the minimizer can be pushed onto the training kernels. In the RKBS built from total variation, the norm is an atomic ℓ^1-type norm instead. A non-training atom can represent the same two observed values with smaller total variation than any combination of the two training atoms. A three-point positive definite kernel already exhibits this geometry.

3 Counterexample

Theorem 1. *There is a finite compact input space X, a continuous positive definite kernel κ on X, and a regularized empirical-risk problem in the RKBS*

$$\mathcal{B} = \{f_\mu = \int_X \kappa(\cdot, t)\, d\mu(t) : \mu \in \mathcal{M}(X)\}$$

with total-variation quotient norm, such that the unique minimizer is not representable by atoms supported on the training inputs. Thus the classical RKHS conclusion "the representer uses only training-site kernels" does not extend to this RKBS setting.

Proof. Let $X = \{-1, 0, 1\}$ with the discrete compact topology. Define the symmetric kernel matrix, in the order $(-1, 0, 1)$, by

$$K = \begin{pmatrix} 1 & 3/5 & 0 \\ 3/5 & 1 & 3/5 \\ 0 & 3/5 & 1 \end{pmatrix}.$$

This is positive definite, since its eigenvalues are

$$1, \qquad 1 - \frac{3\sqrt{2}}{5}, \qquad 1 + \frac{3\sqrt{2}}{5},$$

all positive. Hence it is a continuous positive definite kernel on the finite space X. Since K is invertible, the map $\mu \mapsto f_\mu$ from $\mathcal{M}(X) \cong \mathbb{R}^3$ to functions on X is injective, so the RKBS norm is exactly the total variation norm of the unique representing measure.

Train only at the two endpoints:

$$x_1 = -1, \qquad x_2 = 1, \qquad y_1 = y_2 = 1,$$

and take the convex coercive loss $L(y, z) = 3|y - z|$. Write a measure as

$$\mu = a\delta_{-1} + m\delta_0 + b\delta_1.$$

The predictions at the two training points are

$$u_1 = f_\mu(-1) = a + \frac{3}{5}m, \qquad u_2 = f_\mu(1) = b + \frac{3}{5}m,$$

and the regularized empirical-risk objective is

$$J(\mu) = \frac{3}{2}\big(|1 - u_1| + |1 - u_2|\big) + |a| + |m| + |b|.$$

First we show that any minimizer must interpolate the two labels. Let

$$g(u_1, u_2) = \inf\{|a| + |m| + |b| : a + \tfrac{3}{5}m = u_1,\ b + \tfrac{3}{5}m = u_2\}$$

be the atomic norm induced on the two observed coordinates. Because the two training atoms give the coordinate vectors $(1, 0)$ and $(0, 1)$, we have

$$g(v_1, v_2) \le |v_1| + |v_2|.$$

For any μ with observed vector $u = (u_1, u_2)$, the triangle inequality for g gives

$$g(u) \ge g(1, 1) - g((1, 1) - u) \ge g(1, 1) - \big(|1 - u_1| + |1 - u_2|\big).$$

Therefore

$$J(\mu) \ge g(1, 1) + \frac{1}{2}\big(|1 - u_1| + |1 - u_2|\big).$$

Equality in the minimum is possible only when $u = (1, 1)$.

It remains to solve the minimum-total-variation interpolation problem. The constraints $u_1 = u_2 = 1$ imply

$$a = 1 - \frac{3}{5}m, \qquad b = 1 - \frac{3}{5}m,$$

so the norm to minimize is the one-variable convex function

$$h(m) = 2\left|1 - \frac{3}{5}m\right| + |m|.$$

For $0 \le m \le 5/3$,

$$h(m) = 2 - \frac{1}{5}m,$$

so h decreases on this interval. For $m \ge 5/3$,

$$h(m) = \frac{11}{5}m - 2,$$

so h increases. For $m \le 0$,

$$h(m) = 2 - \frac{11}{5}m,$$

so h again increases as one moves away from zero. Hence the unique minimum is attained at

$$m = \frac{5}{3}, \qquad a = b = 0.$$

Thus the unique minimizer is

$$\mu_* = \frac{5}{3}\delta_0, \qquad f_{\mu_*} = \frac{5}{3}\,\kappa(\cdot, 0).$$

It uses one atom, but that atom is at 0, not at either training input -1 or 1.

Finally, a measure supported only on the training points has $m = 0$. If it interpolates, it must be $\delta_{-1} + \delta_1$, with norm 2, whereas $\|\mu_*\|_{\mathrm{TV}} = 5/3$. Non-interpolating training-supported measures cannot minimize by the preceding lower bound. Hence no minimizer is representable using only training-site atoms. □

4 Verification notes

The proof is exact and finite dimensional. The included checker script in `code/` verifies the positive definiteness of the kernel matrix and symbolically minimizes the one-variable ℓ^1 interpolation objective.

5 Novelty and limitations

Local run indexes were searched for arXiv:2109.09710, RKBS representer theorem, training-site atoms, and related neural-network RKBS terms. No exact prior packet was found. Bounded web searches on June 28, 2026 for RKBS representer atoms at training-set points and arXiv:2109.09710 found no separate known answer beyond the source arXiv record.

This packet does not contradict Theorem 3.9 of [1]. Rather, it proves that the theorem cannot be strengthened, in general, to recover the classical RKHS location conclusion that the atoms occur at the training inputs. The construction is finite and kernel-based, not a ReLU/Radon-transform example.

6 Human-review recommendation

The only substantive review point is whether this finite positive definite kernel RKBS is considered an admissible instance of the paper's RKBS framework. It is a compact finite version of the RBF/kernel-mean-embedding example, and all hypotheses used in the representer-theorem discussion are satisfied.

References

[1] F. Bartolucci, E. De Vito, L. Rosasco, and S. Vigogna, *Understanding neural networks with reproducing kernel Banach spaces*, arXiv:2109.09710v2, 2021. `https://arxiv.org/abs/2109.09710`.